

MATH3303 – Lecture 29

Principal Ideal Domains and the Ascending Chain Condition

Review Notes

Reference: Lee, Abstract Algebra, Section 10.3

Overview

This lecture introduces the next class of structured integral domain: the *principal ideal domain* (PID).

- **Definition of a PID**, with examples (fields K , $\mathbb{Z}$, $K[x]$) and nonexamples ($\mathbb{Z}[x]$, $K[x, y]$, and $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$ which is a PID but not Euclidean)
- **Every Euclidean domain is a PID**
- **Ascending Chain Condition (ACC)** in a PID

Connection to Lecture 28. Lecture 28 finished setting up the divisibility theory in a Euclidean domain (gcd, associates, Euclidean algorithm). The transition here is conceptual: PIDs keep the property “every ideal is principal” but drop the explicit Euclidean function. The Ascending Chain Condition then plays the role of ε in finiteness arguments.

1 Principal Ideal Domains

Definition 1.1: Principal ideal domain (from lecture; cf. Lee, Def. 10.9)

Let R be an integral domain. Then R is said to be a *principal ideal domain* (PID) if every ideal of R is principal.

Example 1.1: Examples of PIDs (from lecture; cf. Lee, Examples 10.12, 10.13)

- (1) Any field K is a PID. K has only two ideals: $\{0\}$ and $K = (1)$. Indeed, if $I \trianglelefteq K$ is nonzero, take $0 \neq a \in I$. Then $a^{-1}a = 1 \in I$, so $I = K$.
- (2) $\mathbb{Z}$ is a PID. (Every ideal of $\mathbb{Z}$ is of the form (n) .)
- (3) $K[x]$ is a PID, where K is a field.

Theorem 1.1: Every Euclidean domain is a PID (from lecture; cf. Lee, Thm. 10.8)

Let (R, ε) be a Euclidean domain. Then R is a PID.

Proof. Let $I \trianglelefteq R$. If $I = \{0\}$, it is principal. Suppose $I \neq \{0\}$. Among all elements of I , choose b so that $\varepsilon(b)$ is minimal (this exists by the well-ordering principle, since ε takes values in $\mathbb{Z}_{\geq 0}$).

Hence $(b) \subseteq I$. **WTS:** $I \subseteq (b)$.

Take any $a \in I$. By the division algorithm, $\exists q, r \in R$ with $a = bq + r$, where $r = 0$ or $\varepsilon(r) < \varepsilon(b)$.

- If $r = 0$, done: $a = bq \in (b)$.
- If $r \neq 0$: $r = a - bq \in I$ (since $a, b \in I$). But $\varepsilon(r) < \varepsilon(b)$, contradicting the minimality of $\varepsilon(b)$.

Hence $I = (b)$, so I is principal. □

2 Nonexamples

Example 2.1: $\mathbb{Z}[x]$ is not a PID (from lecture; cf. Lee, Example 10.14)

The ideal $(x, p) \trianglelefteq \mathbb{Z}[x]$ is *not* principal whenever p is a prime number.

Proof. Suppose for contradiction $(x, p) = (f(x))$ for some $f(x) \in \mathbb{Z}[x]$. Then $p \in (f(x))$, so $f(x) \mid p$ in $\mathbb{Z}[x]$. Hence $f(x)$ is a constant in $\{\pm 1, \pm p\}$.

- $(\pm 1) = \mathbb{Z}[x] \neq (x, p)$, since $1 \notin (x, p)$ (every element of (x, p) has constant term divisible by p).
- $(\pm p) = p\mathbb{Z}[x]$ does not contain x . Contradiction.

Hence $\mathbb{Z}[x]$ is not a PID, and in particular not a Euclidean domain.

Example 2.2: $K[x, y]$ is not a PID (from lecture)

Let K be a field. The ideal $(x, y) \trianglelefteq K[x, y]$ is not principal.

Proof. Suppose for contradiction $(x, y) = (h)$ for some $h \in K[x, y]$. Then $h \mid x$ and $h \mid y$. As x and y are distinct monomials with no common (non-unit) factor, h must be a unit (i.e. a nonzero element of K). But then $(h) = K[x, y]$, so $1 \in (x, y)$. This is a contradiction: every polynomial in (x, y) has zero constant term.

Example 2.3: $\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ is a PID but not Euclidean (from lecture)

$\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ is a PID but not a Euclidean domain. (See the problem sheet.)

This is the standard witness that the inclusion $\{\text{Euclidean}\} \subsetneq \{\text{PIDs}\}$ is strict.

3 The Ascending Chain Condition

Theorem 3.1: Ascending Chain Condition (ACC) (from lecture; cf. Lee, Thm. 10.9)

Let R be a PID. Suppose R has ideals $(I_k)_{k \in \mathbb{N}}$ with

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

Then there exists $N \in \mathbb{N}$ such that $I_k = I_N$ for all $k \geq N$.

Remark 3.1: Concrete chain in $\mathbb{Z}$ (from lecture)

Take $R = \mathbb{Z}$:

$$120 = 2 \times 60 = 2 \times 2 \times 30 = 2 \times 2 \times 2 \times 15 = 2 \times 2 \times 2 \times 3 \times 5,$$

giving

$$(120) \subsetneq (60) \subsetneq (30) \subsetneq (15) \subsetneq (5) \subsetneq (1).$$

In a PID, the ACC produces a (finite) prime factorisation. This will be made precise when we prove every PID is a UFD (Lectures 30–31).

Proof of Theorem 3.1. Set $I := \bigcup_{k=1}^{\infty} I_k$.

Check that $I \leq R$.

- $0 \in I_1 \subseteq I$.
- For $a, b \in I$, say $a \in I_k$ and $b \in I_l$. Let $m = \max(k, l)$. Then $a, b \in I_m$, so $a - b \in I_m \subseteq I$.
- For $a \in I$, say $a \in I_k$, and $r \in R$: $ra \in I_k \subseteq I$.

Hence I is an ideal.

Since R is a PID, $I = (x)$ for some $x \in R$. Since $x \in \bigcup_{k=1}^{\infty} I_k$, $\exists N \in \mathbb{N}$ with $x \in I_N$. Then

$$I = (x) \subseteq I_N \subseteq I_k \subseteq I, \quad \forall k \geq N.$$

Hence $I_k = I_N$ for all $k \geq N$. □

Remark 3.2: Looking ahead (from lecture)

The next lecture uses the ACC and the contrast between *prime* and *irreducible* elements to start narrowing the gap between PIDs and the desirable property of unique factorisation.

Checklist of Skills

- ☐ State the definition of a principal ideal domain.
- ☐ Cite the basic examples: fields K , $\mathbb{Z}$, and $K[x]$.
- ☐ State and prove that every Euclidean domain is a PID.
- ☐ Prove that $\mathbb{Z}[x]$ is not a PID by exhibiting a non-principal ideal (x, p) .
- ☐ Prove that $K[x, y]$ is not a PID by exhibiting the non-principal ideal (x, y) .
- ☐ Quote the fact that $\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ is a PID but not Euclidean.
- ☐ State and prove the Ascending Chain Condition for PIDs.